

AOS-CX 10.16 Monitoring Guide

8100, 83xx, 8360, 9300S, 10000 Switch Series



**Hewlett Packard
Enterprise**

Published: May 2025

Version: 1

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Chapter 1

About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Queue monitoring

Overview

AOS-CX provides multiple features that assist with monitoring queuing behavior such as Queue Monitoring and an extension of it – Congestion Event Detection.

Queue monitoring

The Queue Monitoring feature provides an ongoing collection of queue depth, queue statistics and storage of those statistics, in a time-series database. The time-series data is available for viewing on-switch and provides a history of what has occurred in each of the monitored queues. Queue depth data, accessible on the switch for reviewing the queue statistics history, helps with identifying network issues or behavior patterns.

Queue statistics history

Queue monitoring stores collected queue depth data in a time-series database within the switch. This data, accessible on the switch for reviewing the queue statistics history, helps with identifying network issues or behavior patterns.

Queue depth

Queue depth is the memory size required to buffer packets in a queue during congestion. The value changes depending on the arrival rate of the packets destined to egress a given queue, and the rate at which that queue is able to transmit packets.

Data retention limits

Data retention limits are enforced within the switch by reducing time precision of the stored data within specific windows of time. The *retention duration* for queue statistics data is derived from the frequency of the polling interval.

Table 1: Queue monitoring data retention schedule

Polling interval	Maximum statistic age
10 seconds	8 hours



The switch has limited storage for retaining information. For longer monitoring durations, use an external monitor with the queue monitoring feature to collect and store data.

Using collected data

Queue monitoring provides the ability to look into switch congestion-related performance issues and identify when problems have occurred. The methods available for reviewing performance issues are:

- Congestion events
- Per-interface congestion histograms
- Per-interface congestion-detailed view

As anomalies are identified, the interface queue that experienced the anomaly is the starting point for identifying the cause of the behavior. Examine traffic patterns for flows competing for network bandwidth to determine if corrective action is needed to address potential problems.

Congestion event detection

The Congestion Event Detection feature is an extension of Queue Monitoring that allows users to configure alert thresholds for queue depth, queue rate, and queue drops. Alerts can be viewed on-switch and can also be sent to the switch event log on a selective basis.

Congestion event configuration

The congestion event detection feature is configured with the threshold condition(s) under which an event occurrence starts, ends, and is measured. Each congestion event profile is comprised of zero or more congestion event entries.

Congestion event profile

A congestion event profile entry has a **Valid** status if all the required fields are specified with values that pass validation rules, otherwise, the congestion event profile entry status shows as **Invalid**. Each congestion event profile entry must specify the following information:

- type
- trigger threshold
- reset threshold
- units for provided values
- retention group
- action (optional)

Using congestion events

A congestion event represents a single occurrence of conditions on an interface queue where a configured threshold was violated. Configured thresholds are specified as part of a congestion event profile and are applied on a per-interface basis along with queue monitoring. When a congestion event occurs, relevant data is collected and stored as an *event occurrence* within the congestion event retention group for later analysis. The switch provides 3 retention groups for storing event occurrences, to be used however desired. One effective method is to treat them as "low", "medium", and "high" severity event occurrence storage. This method allows to quickly check the switch for specific severity events using the **show congestion-event retention-group <NUMBER>** command.



Once a congestion event retention group has reached the limit of 1000 stored events, the oldest completed event occurrence in the group is discarded in order to store a new event.

Congestion event detection commands

apply congestion-event profile

```
apply congestion-event profile [<NAME>]
no apply congestion-event profile [<NAME>]
```


Description

Applies a congestion event profile to an interface.

The **no** form removes the congestion event profile from the interface.

Parameter	Description
<code><NAME></code>	Specifies the name of a congestion event profile. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Usage

In order for the congestion event profile entry associated with an applied congestion event profile to be effective, the interface needs to be configured for **queue-monitor**. Once the queue statistics are being monitored, apply the congestion event profile to the interface to begin the process of actively checking the monitored queue statistics for congestion events based on the thresholds configured in the applied congestion event profile. User-specified congestion event profiles and profile entries may be modified and deleted while applied on an interface. If a congestion event profile or entry is modified while applied on an interface, any ongoing event occurrences will end immediately.

Examples

Applying congestion event profile, **prof1**, to an interface where the queues on the interface have a monitoring source:

```
switch(config-if)# apply congestion-event profile prof1
```

Applying non-existent congestion event profile, **prof22**, to an interface:

```
switch(config-if)# apply congestion-event profile prof22  
Profile prof22 does not exist.
```

Applying congestion event profile, **prof1**, to an interface where the queues on the interface do not have a queue data monitoring source. Without a source for queue statistics, no congestion event threshold checking will be performed.

```
switch(config-if)# apply congestion-event profile prof1  
Warning: Queue monitoring must also be enabled using the 'queue-monitor' command  
for congestion events to be detected.
```

Removing any congestion event profile from an interface.

```
switch(config-if)# no apply congestion-event profile
```

Removing congestion event profile, **prof1**, from an interface:

```
switch(config-if)# no apply congestion-event profile prof1
```

Removing congestion event profile, **prof2**, from an interface where congestion event profile, **prof1**, is currently applied.

```
switch(config-if)# no apply congestion-event profile prof2
The profile to remove does not match the currently configured profile.
```

Command History

Release	Modification
10.16	Command introduced

Command Information

Platforms	Command context	Authority
8325 8325H 8325P 10000	config-if	Administrators or local user group members with execution rights for this command.

congestion-event profile

```
congestion-event profile <NAME>
no congestion-event profile <NAME>
```

Description

Creates a new congestion event profile and switches to the **config-cng-prof** context for the profile. The no form deletes the profile and removes it from all applied interfaces.



If the specified profile already exists, this command switches to the **config-cng-prof** context for the named profile.

Parameter	Description
<name>	Name of the congestion event profile, up to 64 characters.

Usage

A congestion event profile contains zero or more congestion event entries that specify the thresholds to use for identifying conditions of interest that should be saved as an event occurrence. Use **show congestion-event profile [NAME]** to view the status of all congestion event profiles or their settings.

Examples

Creating a congestion event profile named **prof1**:

```
switch(config)# congestion-event profile prof1
```

Deleting a congestion event profile named **prof1**:

```
switch(config)# no congestion-event profile prof1
```

Creating a congestion event profile with an invalid name:

```
switch(config)# congestion-event profile prof##
Invalid name. Please enter a string of up to 64 alphanumeric,
underscore, hyphen, and period characters.
```

Command History

Release	Modification
10.16	Command introduced

Command Information

Platforms	Command context	Authority
8325 8325H 8325P 10000	config	Administrators or local user group members with execution rights for this command.

event-config-id

```
event-config-id {<event-type>} | {trigger <units> <value>} | [reset <units> <value>] |
[retention-group <retention-group>] | [action <action> <action-parameter>]
no event-config-id [<event-type>] [trigger <units> <value>] | [reset <units> <value>] |
[retention-group <retention-group>] | [action <action> <action-parameter>]
```

Description

Creates a new congestion event profile entry associated with the current congestion event profile.

The **no** form deletes the congestion event entry from the congestion event profile.

Parameter	Description
event-type	Specifies the type of congestion event. Congestion event types include the following: ▪ queue-depth ▪ queue-drop ▪ queue-tx-rate
[trigger <units> <value>]	Specifies the units and value for the queue statistic trigger threshold.
[reset <units> <value>]	Specifies the units and value for the queue statistic reset threshold. If not provided, the reset value and units will be assigned the same values as the trigger units and value. Note: Units for the trigger and reset values must be

Parameter	Description
	equal to or below the trigger value to have a valid congestion event entry.
[retention-group <retention-group-id>]	(Optional) Specifies the retention group used to store congestion event occurrences for the congestion event entry. If no retention group is provided, congestion event retention group 1 will be used.
[action <action> <action-parameter>]	Specifies an action to take when a congestion event occurs. Available actions and their corresponding parameters: ■ eventlog: Logs an event to the system eventlog which may be seen with show events. The action parameter specifies at which level to log a message: ■ info (informational level) ■ warn (warning level) ■ crit (critical level) Note: If no action is specified, no action is taken with event occurrences.

Examples

Configuring congestion event entries for queue-depth:

```
switch(config-cng-prof) # queue-depth trigger kbytes 3000 reset kbytes 2000
retention-group 2 action eventlog crit
switch(config-cng-prof) # 7 queue-depth trigger kbytes 5000 reset kbytes 1000
```

Removing the congestion event entry with ID 7:

```
switch(config-cng-prof) # no 7
```

Configuring congestion event entries for queue-drops:

```
switch(config-cng-prof) # 2 queue-drops trigger packets 20 reset packets 10
retention-group 3
```

Removing the congestion event entry with ID 2:

```
switch(config-cng-prof) # no 2
```

Configuring congestion event entries for queue-tx-rate:

```
switch(config-cng-prof) # 5 queue-tx-rate trigger kbps 200000 retention-group 1
action eventlog info
```

```
switch(config-cng-prof) # queue-tx-rate trigger pps 7800 reset pps 5000 retention-group 3
```

Removing the congestion event entry with ID 5:

```
switch(config-cng-prof) # no 5
```

Command History

Release	Modification
10.16	Command introduced

Command Information

Platforms	Command context	Authority
8100 8325 8325H 8325P 8360 9300 10000	config	Administrators or local user group members with execution rights for this command.

show congestion-event profile

```
show congestion-event profile [<NAME>]
```

Description

Shows the status of all congestion event profiles, or the settings of a specific congestion event profile.

Parameter	Description
<NAME>	Specifies the name of a congestion event profile. Range: 1 to 64 alphanumeric characters, including period (.), underscore (_), and hyphen (-).

Usage

The status of each congestion event entry can be one of the following:

- **Valid** - the congestion event entry is valid.
- **Invalid** - the congestion event entry is invalid.

Use the **show congestion-event profile <NAME>** command to see details about each congestion event entry and reasons for rejection.

Examples

Showing the status of all congestion event profiles. The numbers provided in **Config Entry Status** field show the count of how many entries are currently in each status.

```
switch# show congestion-event profile
Name          Config Entry Status
-----
prof1         2 Valid
prof2         1 Valid, 1 Invalid
prof3         -
```

Showing the status of congestion event profile **prof1**:

```
switch# show congestion-event profile prof1
Profile Name      : prof1
Applied Interfaces : 1/1/1-1/1/3
Retention
ID State  Event Type          Trigger          Reset          Group Action
--
*1 Invalid Queue Depth      -                -                -      eventlog/crit
2 Valid   Queue Depth      12345 kbytes    12345 kbytes    2      -
3 Valid   Queue Drops      1234 pkts      123 pkts        1      -
*4 Invalid Queue Drops      600 pkts       50 pkts         -      eventlog/info
7 Valid   Average Queue Tx Rate 75% line rate 50% line rate 1      eventlog/error
8 Valid   Average Queue Tx Rate 12345 pps      1234 pps        2      -
9 Valid   Average Queue Tx Rate 54321 kbps     54321 kbps      2      -

*1: Missing value: trigger, reset, retention group.
*4: Missing value: retention group.
```

Command History

Release	Modification
10.16	Command introduced

Command Information

Platforms	Command context	Authority
8325 8325H 8325P 10000	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show interface congestion-event

```
show interface [<IFNAME>] congestion-event
```

Description

Displays the congestion event status on an interface.

Parameter	Description
<IFNAME>	Specifies the name of the interface.

Examples

Displaying the congestion event status on an interface that does not have a congestion event profile applied:

```
switch# show interface 1/1/1 congestion-event
Interface 1/1/1 does not have a congestion event profile applied.
```

Displaying the congestion event status on an interface that is not currently monitoring queue statistics and has a congestion event profile applied:

```
switch# show interface 1/1/1 congestion-event
Congestion Event Profile : cng_event_prof1
Queue 0-7
Aggregation Interval :
Data Source          :
Event Types          :
Status               : Not Applied (No queue monitoring source for event data)
```

Displaying the congestion event status on an interface that is down:

```
switch# show interface 1/1/1 congestion-event
Congestion Event Profile : cng_event_prof1
Queue 0-7
Aggregation Interval :
Data Source          :
Event Types          :
Status               : Not Applied (Interface is down)
```

Displaying the congestion event status on an interface where the applied congestion event profile contains only invalid entries:

```
switch# show interface 1/1/1 congestion-event
Congestion Event Profile : cng_event_prof1
Queue 0-7
Aggregation Interval :
Data Source          :
Event Types          :
Status               : Not Applied (2 invalid entries in congestion-event profile)
```

Displaying the congestion event status on an interface that is actively collecting queue statistics with queue monitoring, but some entries in the applied profile are invalid:

```
switch# show interface 1/1/1 congestion-event
Congestion Event Profile : cng_event_prof1
Queue 0-7
Aggregation Interval : 10s
Data Source          : queue-monitor
Event Types          : queue-depth, queue-drops, queue-tx-rate
```

```
Status                : Partially Applied (1 invalid entry in congestion-event
profile)
```

Displaying the congestion event status on an interface that is actively collecting queue statistics with queue monitoring:

```
switch# show interface 1/1/1 congestion-event
Congestion Event Profile : cng_event_prof1
Queue 0-7
Aggregation Interval : 10s
Data Source          : queue-monitor
Event Types          : queue-depth, queue-drops, queue-tx-rate
Status               : Applied
```

Displaying congestion event status on all interfaces:

```
switch# show interface congestion-event
Interface 1/1/1
Congestion Event Profile : cng_event_prof1
Queue 0-7
Aggregation Interval : 10s
Data Source          : queue-monitor
Event Types          : queue-depth, queue-drops, queue-tx-rate
Status               : Applied
Interface 1/1/2
Congestion Event Profile : cng_event_prof1
Queue 0-7
Aggregation Interval : 10s
Data Source          : queue-monitor
Event Types          : queue-depth, queue-drops, queue-tx-rate
Status               : Applied
```

Command History

Release	Modification
10.16	Command introduced

Command Information

Platforms	Command context	Authority
8325 8325H 8325P 10000	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show congestion-event

```
show congestion-event
  filter [since <time-value> | {seconds|minutes|hours|days}] | [state
{ongoing|complete}] | [interface <ifname|ifrange>] | [queue <queues>][retention-group
<retention-group>] | [type {queue-tx-rate|queue-drops|queue-depth}]
```


Description

Shows congestion events that have occurred.

Parameter	Description
[since <time-value> {seconds minutes hours days}]	Filters the output by only displaying event occurrences that were updated equal to or later than the time given, where <time-value> and {seconds minutes hours days} specifies how far backward to look for events, relative to the current time.
[state {ongoing complete}]	Filters event occurrences based on whether the event occurrence is ongoing or complete (ended).
[interface <ifname ifrange>]	Specifies the name of the interface or range of interfaces.
[queue <queues>]	Filters event occurrences based on the queue(s) where the event occurred. Range: 0 to 7.
[retention-group <retention-group>]	Filters event occurrences based on the retention group where they are stored.
[type {queue-tx-rate queue-drops queue-depth}]	Filters event occurrences based on the type of event that occurred.

Usage

For more detailed information on an event occurrence, use the **show congestion-event <id>** command.

Examples

Displaying all congestion event occurrences, but none matching the provided filters have occurred yet:

```
8325(config-cng-prof)# show congestion-event
Matching Event Count: 0
```
Display all of the congestion event occurrences.
```
switch# show congestion-event
Matching Event Count: 4

O - Ongoing
C - Complete

ID      Start time      Duration      Type                               Interface  Queue  Value
-----
124 10 seconds ago  10s (O)      Queue Depth                        1/1/1     3
123456 kbytes
122 20 minutes ago  20m (O)      Average Queue Tx Rate             1/1/2     1      9283
kbps
```

```

123 3 minutes ago 50m (C) Queue Depth 1/1/4 6
728394 kbytes
121 1 hour ago 2m (C) Queue Drops 1/1/7 7 6233
pkts

```

Displaying all congestion event occurrences that were updated in the last 5 minutes on **interfaces 1/1/1 and 1/1/4**:

```

switch# show congestion-event filter since 5 minutes interface 1/1/1,1/1/4
Matching Event Count: 2

O - Ongoing
C - Complete

ID   Start time      Duration  Type           Interface Queue  Value
-----
--
124  10 seconds ago  10s (O)   Queue Depth    1/1/1   3     123456 kbytes
123  2 days ago     2m (C)   Queue Depth    1/1/4   6     728394 kbytes

```

Displaying all congestion event occurrences that are still ongoing:

```

switch# show congestion-event filter state ongoing
Matching Event Count: 2

O - Ongoing
C - Complete

ID   Start time      Duration  Type           Interface Queue  Value
-----
--
124  10 seconds ago  10s (O)   Queue Depth    1/1/1   3     123456
kbytes
122  20 minutes ago  20m (O)   Average Queue Tx Rate 1/1/2   1     9283 kbps

```

Displaying all congestion event occurrences on **interface 1/1/4**:

```

switch# show congestion-event filter interface 1/1/4
Matching Event Count: 1

O - Ongoing
C - Complete

ID   Start time      Duration  Type           Interface Queue  Value
-----
--
123  3 hours ago     10s (C)   Queue Depth    1/1/4   6     728394 kbytes

```

Displaying all congestion event occurrences on interface 1/1/1 and queue 3:

```

switch# show congestion-event filter interface 1/1/1 queue 3
Matching Event Count:1

O - Ongoing
C - Complete

```

ID	Start time	Duration	Type	Interface	Queue	Value
124	10 minutes ago	10m (0)	Queue Depth	1/1/1	3	123456 kbytes

Displaying all congestion event occurrences stored in **retention group 3** and type **queue-depth**:

```
switch# show congestion-event filter retention-group 3 type queue-depth
Matching Event Count: 1

O - Ongoing
C - Complete
```

ID	Start time	Duration	Type	Interface	Queue	Value
124	10 seconds ago	10s (0)	Queue Depth	1/1/1	3	123456 kbytes

If the congestion event entry includes an action to log to the system event log, display the event log to view start and end information for logged occurrences. Displaying event log:

```
switch# show events
2024-11-01T23:43:11.778434+00:00 8325 switchd_agent[3436]: Event|1502|LOG_
INFO|AMM|1/1|Congestion event 123: start - interface 1/1/1 queue 3 - queue-depth:
5100 kbytes
2024-11-01T23:43:21.999842+00:00 8325 switchd_agent[3436]: Event|1503|LOG_
INFO|AMM|1/1|Congestion event 123: end - interface 1/1/1 queue 3 - lasted 10
seconds with peak value 65656 kbytes
```

Command History

Release	Modification
10.16	Command introduced

Command Information

Platforms	Command context	Authority
8325 8325H 8325P 10000	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show congestion-event <id>

```
show congestion-event <id>
```

Description

Shows detailed output of a single congestion event occurrence.

Parameter	Description
<id>	Specifies the congestion event occurrence to display.

Examples

Displaying details of an ongoing Average Queue Tx Rate congestion event occurrence:

```
switch# show congestion-event 123
Congestion Event Occurrence 123

Event Configuration
  Data Source      : queue-monitor
  Aggregation Interval : 10s
  Type             : Average Queue Tx Rate
  Trigger          : 12345 pps
  Reset            : 1234 pps
  Retention Group   : 2

Event Data
  Interface        : 1/1/1
  Queue            : 3
  Start Time       : 10 minutes ago (2024-03-09 10:02:24)
  End Time         : N/A (Ongoing Event)
  Duration         : 10m
  Peak Time        : 2024-03-09 10:11:34
  Peak             : 123456 pps
```

Displaying details of a completed Queue Depth congestion event occurrence:

```
switch# show congestion-event 124
Congestion Event Occurrence 124

Event Configuration
  Data Source      : queue-monitor
  Aggregation Interval : 10s
  Type             : Queue Depth
  Trigger          : 12345 kbytes
  Reset            : 1234 kbytes
  Retention Group   : 1

Event Data
  Interface        : 1/1/2
  Queue            : 5
  Start Time       : 20 seconds ago (2024-03-09 10:02:24)
  End Time         : 10 seconds ago (2024-03-09 10:02:34)
  Duration         : 10s
  Peak Time        : 2024-03-09 10:02:29
  Peak             : 123456 kbytes
```

Displaying details of a completed Queue Drop congestion event occurrence:

```
switch# show congestion-event 125
Congestion Event Occurrence 125

Event Configuration
  Data Source      : queue-monitor
```

```
Aggregation Interval : 10s
Type                  : Queue Drops
Trigger               : 12345 packets
Reset                 : 1234 packets
Retention Group       : 1

Event Data
Interface             : 1/1/2
Queue                 : 5
Start Time             : 7 hours ago (2024-03-09 10:05:24)
End Time               : 7 hours ago (2024-03-09 10:12:44)
Duration              : 7m
Peak Time              : 2024-03-09 10:05:29
Peak                   : 123456 packets
Sum                    : 234567 packets
```

Displaying the details of congestion event occurrence 125 when this congestion event occurrence ID is invalid:

```
switch# show congestion-event 125
Congestion event occurrence 125 is not valid.
```



An ID is invalid if the occurrence ID is not being used yet. If the ID *is* valid and displaying "invalid", the congestion event retention group deleted the occurrence to make room for new occurrences.

Command History

Release	Modification
10.16	Command introduced

Command Information

Platforms	Command context	Authority
8325 8325H 8325P 10000	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.